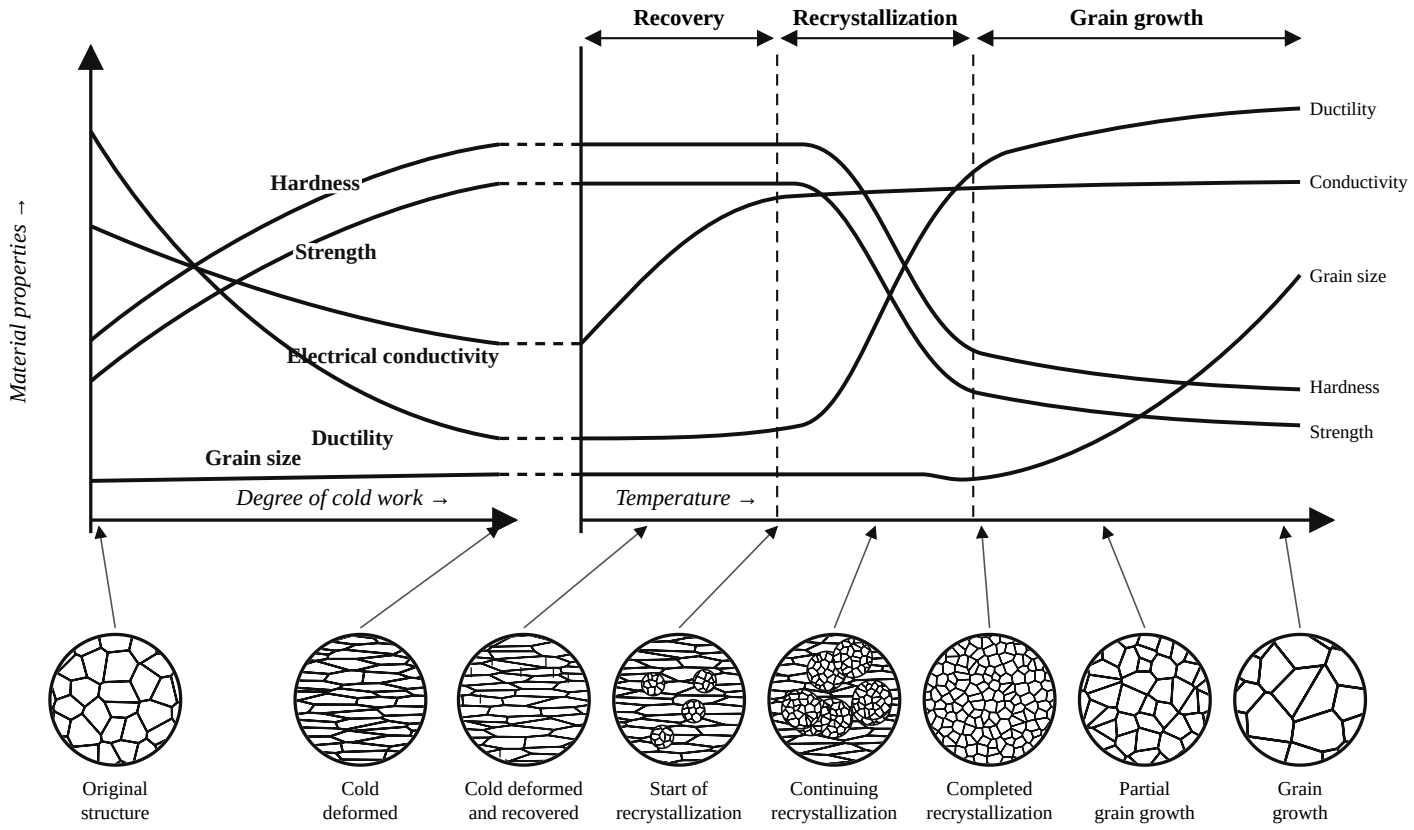
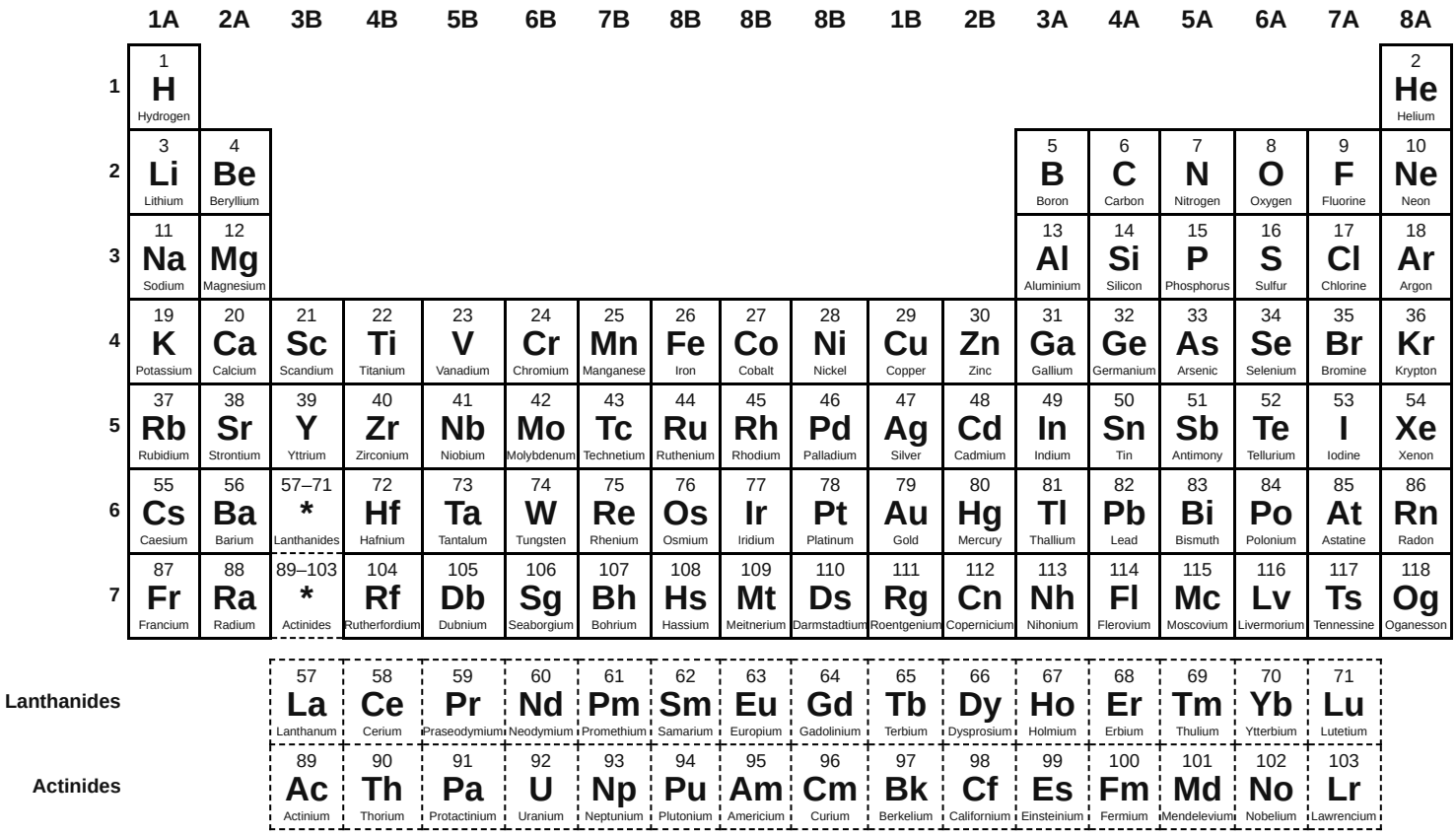
Effect of Cold Work and Annealing on Material Properties



Effect of cold work and annealing on material properties

The diagram shows how a metal's properties change during **cold working** and then during **heating (annealing)**. As cold work increases, the grains become **elongated** and dislocations build up, so **hardness and strength increase** (strain hardening) while **ductility and electrical conductivity decrease**. On annealing, three stages occur: **(1) Recovery** – internal stresses are relieved and conductivity rises, but hardness, strength and grain shape barely change; **(2) Recrystallization** – new strain-free equiaxed grains replace the deformed ones, so hardness and strength drop sharply and ductility is restored; **(3) Grain growth** – at higher temperatures the new grains grow larger, further lowering strength and raising ductility.

Periodic Table – Groups



1A–2A & 3A–8A = A groups (main group) · 3B → 2B = B groups (transition metals) · dashed boxes = inner transition metals

In the periodic table, the **vertical columns are called groups**. They are divided into two types: **A groups** (main group elements) and **B groups** (transition metals).

Special names of the A groups (main group elements)

- **Group 1A:** Alkali metals (except hydrogen)
- **Group 2A:** Alkaline earth metals
- **Group 3A:** Earth metals (Boron group)
- **Group 4A:** Carbon group
- **Group 5A:** Nitrogen group
- **Group 6A:** Oxygen group (Chalcogens)
- **Group 7A:** Halogens
- **Group 8A:** Noble gases (Inert gases)

Special names of the B groups (transition metals)

- **Between 3B and 2B (10 columns):** called **transition metals**, and all of them are metals.
- **Lanthanides and Actinides** are located at the bottom of the table and are called **inner transition metals**.

Types of Atomic Bonds

Three types

- **Ionic bond**
- **Covalent bond**
- **Metallic bond**

Metallic bond (brief)

A metallic bond forms between metal atoms that give up their valence electrons. These electrons become a **"sea" (cloud) of free electrons** shared by all the atoms, and the positive metal ions are held together by their attraction to this electron cloud. The bond is **non-directional**, which explains why metals are **good conductors of electricity and heat**, **ductile** and **malleable**, and have a shiny lustre.